A3-PS019-TC358

Personal-Data Discovery and Privacy Evidence Inventory

Team members:

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Problem Description

- Personal-data fields may be spread across CSV spreadsheets and SQL database tables.
- Teams may lack a single inventory linking fields to declared business purpose and retention.
- Missing purpose or retention metadata can make review and evidence collection difficult.
- This project proposes a configurable scanner that records candidate fields and rule evidence, then flags gaps.
- Findings are technical candidates for human verification, not legal or compliance conclusions.

Background

Personal-data discovery is a technical first step in building an evidence inventory. Field names can provide clues, while sample values can provide pattern-based evidence. Neither method is perfect: names may be ambiguous, values may be masked or inconsistent, and unrelated fields may share similar formats. The prototype records why a field was flagged and leaves final confirmation to a human reviewer. The demonstration uses synthetic data to avoid exposing real personal information.

Literature & RFC Review

The project uses the MeitY Data Protection Framework as the primary policy reference identified for this challenge. The technical design follows evidence-traceability and data-minimisation principles: inspect configured sources and bounded samples, record the rule that triggered a candidate, and separate technical observations from legal interpretation. The team should cite only documents actually reviewed and confirm any additional standards requested by the mentor. This prototype does not determine lawful basis or certify compliance.

Proposed Solution

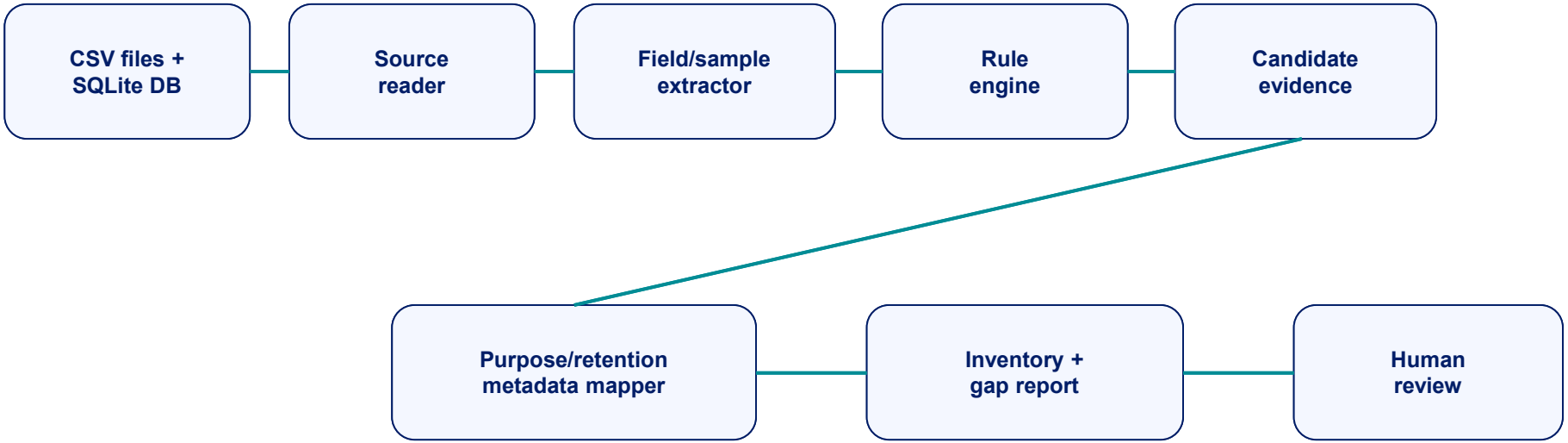
1. Source ingestion: read configured CSV files and inspect tables in a SQLite database.
2. Field and sample extraction: collect field names and a bounded sample of values.
3. Configurable detection: apply rules for candidate classes PERSON_NAME, EMAIL, PHONE, GOVERNMENT_ID, and ADDRESS.
4. Evidence recording: save source, table/file, field, candidate class, and matching rule evidence.
5. Metadata mapping: attach declared business purpose and retention details; flag missing values.
6. Reporting and review: export inventory, metadata gaps, and per-class precision/recall for human review.

Solution Optimization

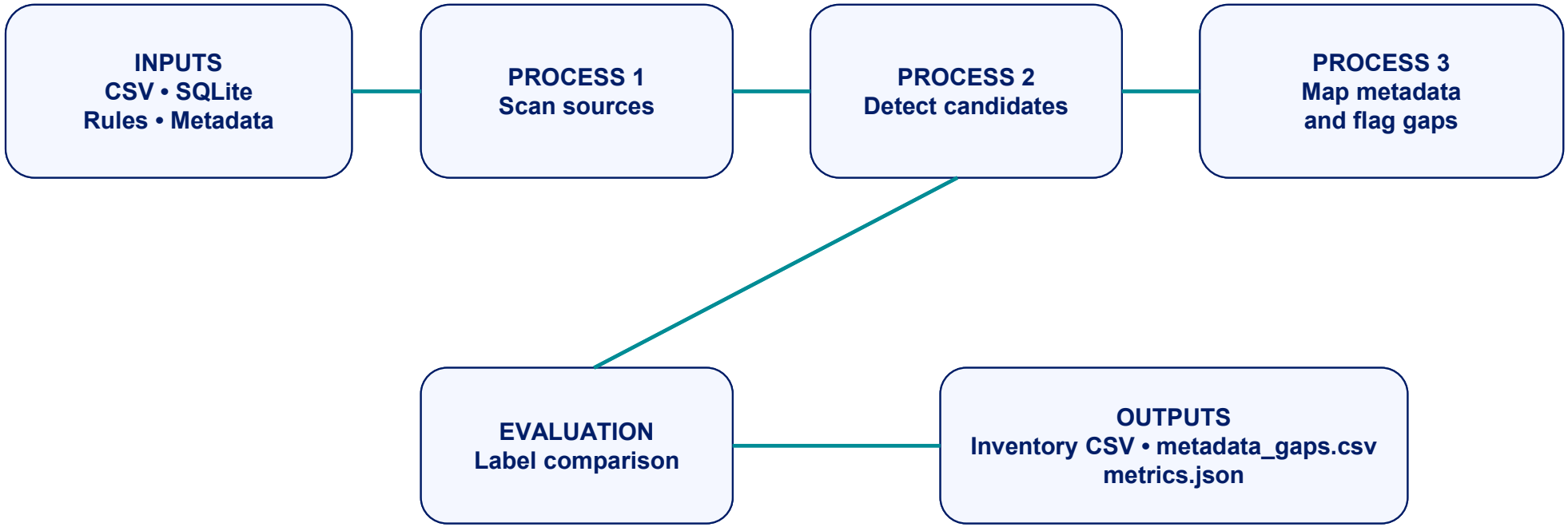
The prototype replaces fragmented manual checks with a repeatable, configurable workflow. Benefits remain qualitative until measured in repeatable tests.

Aspect	Before	Proposed approach
Discovery	Manual checks in separate files	Repeatable CSV + SQLite scan
Rules	Ad hoc identification	Configurable detection rules
Metadata	Notes may be incomplete	Purpose/retention mapping + gap flags
Evidence	Hard to reproduce	Source, field and rule evidence exported
Evaluation	No consistent scorecard	Per-class precision and recall

Solution Architecture & Design



Data Flow Diagram



Thank You for Your Attention!

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